Nicholas Yap

Mechanical Engineer, EIT

780-266-7748 nicholas.yap28@gmail.com linkedin.com/in/nicholas-yap28 Personal Portfolio

Education

University of British Columbia **Sep. 2021 – May 2026**

Bachelor of Applied Science in Mechanical Engineering, Mechatronics Specialization — CGPA: 87.6% *Vancouver, BC*

Relevant Coursework

- Sensors and Actuators
- Mechatronics Systems
- Control Systems
- Product Design
- Industrial Robotics
- Electromechanics
- Digital Systems
- Electronic Circuits

Experience

Ted Rogers Centre for Heart Research **May 2025 – Aug. 2025**

Summer Research Student – Cardiac Surgery Department *Toronto, ON*

- Configured a secure REDCap data collection framework and developed MATLAB scripts to clean and restructure aortic biomechanics research datasets, improving data consistency and management.

The Dow Chemical Company **Feb. 2024 – Aug. 2024**

Maintenance/Reliability Engineering Co-op *Fort Saskatchewan, AB*

- Programmed preset temperature-cycle recipes for an industrial oven, automating heating, holding, and cooling sequences to improve flexibility across shutdown procedures.
- Investigated pressure-equipment die wear and scratches, consulted materials engineers, and sourced stainless steel cleaning scrapers; revised cleaning procedures and developed operator training materials.
- Coordinated 15+ end-to-end Management of Change (MOC) processes for equipment installations, replacements, and upgrades, reviewing engineering drawings, updating P&IDs and maintenance procedures, coordinating vendor requirements and approvals, and tracking implementation.

The Dow Chemical Company **Sep. 2023 – Feb. 2024**

Fixed Asset Integrity Engineering Co-op *Fort Saskatchewan, AB*

- Coordinated with operations teams across 5+ plant facilities to update corrosion-control, operating-window, and ABSA regulatory documentation supporting continued site operation.

UBC Uncrewed Aircraft Systems **Sep. 2022 – Sep. 2026**

Senior Aircraft and Administrative Member *Vancouver, BC*

- Developed bills of materials and spreadsheets to calculate component moments and assess weight distribution for two fixed-wing aircraft, informing component placement and fuselage positioning; designed CAD components and supported fabrication and integration.

Projects

Soft-Robotics Claw Prototype | *Onshape, TPU, PLA* **Apr. 2026**

- Created 5+ CAD iterations of a logarithmic-spiral soft robotic claw, prototyping TPU flexible joints and PLA hinges to evaluate geometry, actuation routing, and structural trade-offs.

Self-Balancing Wheeled Bipedal Robot | *Onshape, ESP-32, C#, BLE* **Feb. 2026**

- Designed a symmetric, modular chassis to align the IMU and centralize mass for stable balancing; iterated through six wheel/rim designs to address gearmotor backlash, incorporating TPU honeycomb tires and protective bumpers.
- Developed a C# WinForms application for wireless BLE control, real-time tilt monitoring, and data logging; checked chassis balance and tested traction across multiple surfaces.

2-Axis Motorized Gantry | *C/C++, MSP430, PID Control* **Dec. 2025**

- Developed embedded firmware with encoder feedback, PID positioning, timers, interrupts, and UART command handling to control stepper and DC motors for repeatable, G-code-like drawing sequences.

Technical Skills

Mechanical Design: SOLIDWORKS, Onshape, GD&T, DFMA, 3D printing and prototyping
Programming/Software: C/C++, C#, Python, MATLAB, WinForms
Embedded Systems/Controls: MSP430 firmware, Arduino, UART, BLE, PID control, PLC ladder logic
Electronics: Electronics debugging, oscilloscope, DMM, function generator, power supply